EXP 7

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clc;
clear;
close all;

% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signals Using MATLAB

%% 1. System Parameters
Fs = 200000;            % Sampling frequency (Hz)
T = 0.05;               % Simulation duration (seconds)
t = 0:1/Fs:T-1/Fs;      % Time vector

Am = 1;                 % Message signal amplitude (V)
fm = 200;               % Message frequency (Hz)
Ac = 1;                 % Carrier amplitude (V)
fc = 5000;              % Carrier frequency (Hz)
kf = 1000;              % Frequency sensitivity (Hz/V)

%% 2. Generate Message Signal
m = Am*cos(2*pi*fm*t);

%% 3. Calculate Frequency Deviation
delta_f = kf*Am;

%% 4. Calculate FM Modulation Index
beta = delta_f/fm;

%% 5. Generate FM Signal
% Instantaneous phase of FM signal
phase_FM = 2*pi*fc*t + 2*pi*kf*cumtrapz(t,m);
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% FM modulated signal
sFM = Ac*cos(phase_FM);

%% 6. Calculate Instantaneous Frequency
% Numerical derivative of instantaneous phase
inst_freq = [diff(phase_FM)*Fs/(2*pi), NaN];

%% 7. Verify Frequency Deviation
f_max = fc + delta_f;
f_min = fc - delta_f;

fprintf('\n===== FM Parameter Calculation =====\n');
fprintf('Carrier Frequency       = %.2f Hz\n',fc);
fprintf('Message Frequency       = %.2f Hz\n',fm);
fprintf('Message Amplitude       = %.2f V\n',Am);
fprintf('Frequency Sensitivity   = %.2f Hz/V\n',kf);
fprintf('Maximum Frequency Dev.  = %.2f Hz\n',delta_f);
fprintf('FM Modulation Index     = %.2f\n',beta);
fprintf('Maximum Instantaneous F = %.2f Hz\n',f_max);
fprintf('Minimum Instantaneous F = %.2f Hz\n',f_min);

%% 8. Plot Message Signal
figure('Name','FM Frequency Deviation and Modulation Index');

subplot(4,1,1);
plot(t,m,'b','LineWidth',1.2);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
grid on;
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%% 9. Plot FM Signal
subplot(4,1,2);
plot(t,sFM,'r');
title('Frequency Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

%% 10. Plot Instantaneous Frequency
subplot(4,1,3);
plot(t,inst_freq,'k','LineWidth',1.1);
hold on;
yline(fc,'b--','Carrier Frequency');
yline(f_max,'r--','Maximum Frequency');
yline(f_min,'g--','Minimum Frequency');
title('Instantaneous Frequency of FM Signal');
xlabel('Time (s)');
ylabel('Frequency (Hz)');
grid on;
ylim([fc-1.5*delta_f fc+1.5*delta_f]);

%% 11. Frequency Spectrum
N = length(sFM);
f = (-floor(N/2):ceil(N/2)-1)*(Fs/N);
S = abs(fftshift(fft(sFM))/N);

subplot(4,1,4);
plot(f/1000,S,'m');
xlim([0 12]);
title('FM Frequency Spectrum');
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xlabel('Frequency (kHz)');
ylabel('Magnitude');
grid on;

%% 12. Verification
fprintf('\n===== Verification =====\n');
fprintf('Theoretical Frequency Deviation = %.2f Hz\n',delta_f);
fprintf('Theoretical Modulation Index    = %.2f\n',beta);
fprintf('Expected Frequency Range        = %.2f to %.2f Hz\n', ...
    f_min,f_max);
disp('FM simulation and verification completed successfully.');
```

